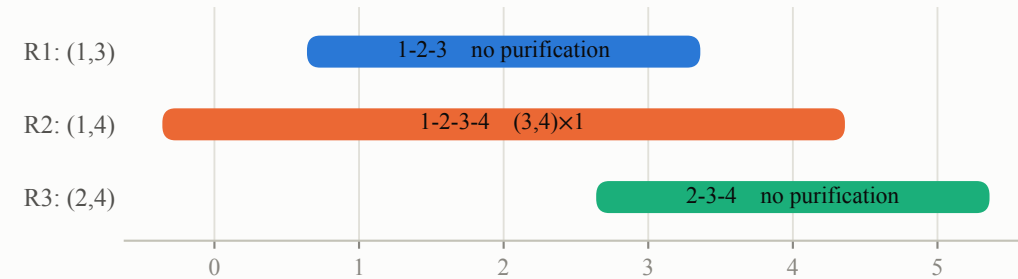


Small-scale optimality validation on a known 4-node chain: WPFA versus the exact optimum

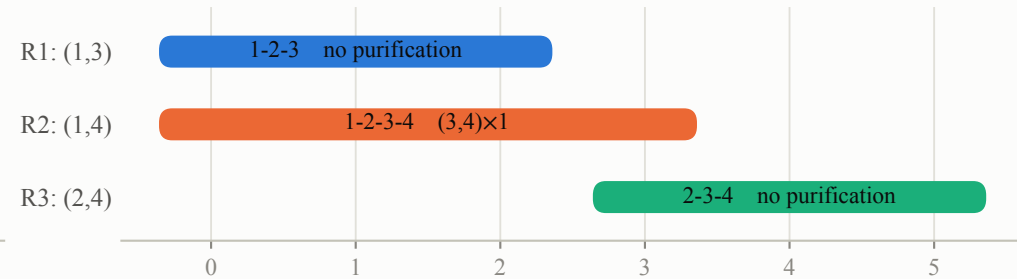
Both methods accept all three requests and select the same paths and the same purification. Expected Werner sum: exact OPT 2.072766, WPFA 2.071154 -- an optimality gap of 0.078%.

Only R2 differs, and only in when its operations are placed; the whole gap comes from that one request's storage time.

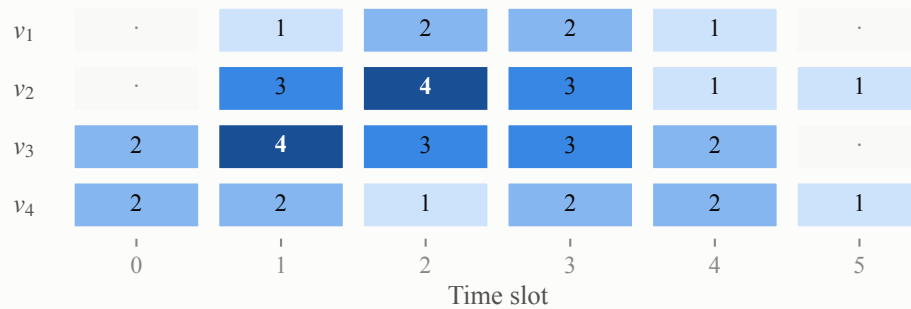
Exact OPT: selected schedule



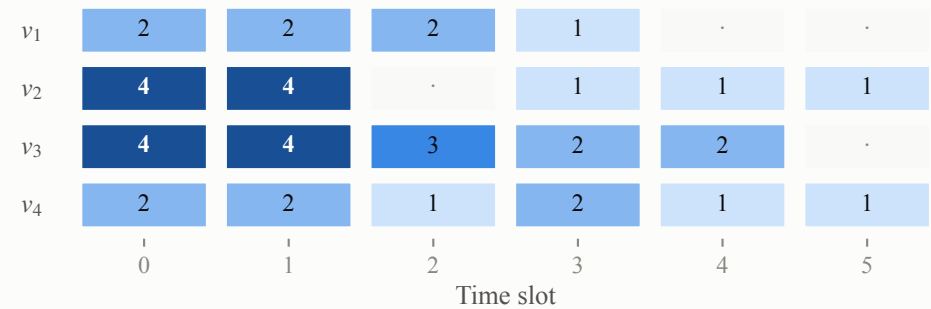
WPFA: selected schedule



Exact OPT: numerology occupancy



WPFA: numerology occupancy



memory cells held $\theta_m(t, v)$: 0 1 2 3 4

capacity 4 per node-slot (bold = saturated); both methods hold the same total, $\sum \theta = 42$

Request	Method	Path	Purification	Slots	F	w	Pr	Pr · w
R1: (1,3)	Exact OPT	1-2-3	no purification	1-3	0.8680	0.8240	0.8988	0.740651
R1: (1,3)	WPFA	1-2-3	no purification	0-2	0.8680	0.8240	0.8988	0.740651
R2: (1,4)	Exact OPT	1-2-3-4	(3,4)×1	0-4	0.8084	0.7446	0.7862	0.585356
R2: (1,4)	WPFA	1-2-3-4	(3,4)×1	0-3	0.8069	0.7425	0.7862	0.583744
R3: (2,4)	Exact OPT	2-3-4	no purification	3-5	0.8731	0.8308	0.8989	0.746759
R3: (2,4)	WPFA	2-3-4	no purification	3-5	0.8731	0.8308	0.8989	0.746759
All three	Exact OPT	3 accepted	exact optimum					2.072766
All three	WPFA	3 accepted	gap 0.078%					2.071154

Exact OPT enumerates every schedule of the discrete-time model on every simple path (1,584 schedules, 900 feasible, 197 branch-and-bound states) with no approximation bucket, candidate cap, or time cutoff.